AppWorld
Mean ± SD

	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
Method	**Stage 0**	**Stage 1**	**Stage 0**	**Stage 1**	**Stage 0**	**Stage 1**
Frozen Base	0.2723 ±0.0316	0.2723 ±0.0316	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3482 ±0.0337
AREW / T3	0.2723 ±0.0316	0.2754 ±0.0282	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3646 ±0.0352
GiGPO	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.3887 ±0.0397
InfoPO	0.2723 ±0.0316	0.2995 ±0.0162	0.2877 ±0.0285	0.3277 ±0.0139	0.3482 ±0.0337	0.4405 ±0.0330
WorldEvolver	0.2723 ±0.0316	0.4646 ±0.0070	0.2877 ±0.0285	0.4732 ±0.0253	0.3482 ±0.0337	0.4800 ±0.0094
PoS / EnvProbe	0.2723 ±0.0316	**0.4800** ±0.0158	0.2877 ±0.0285	**0.5062** ±0.0289	0.3482 ±0.0337	**0.6430** ±0.0104
Step-DPO	0.2723 ±0.0316	0.4118 ±0.0455	0.2877 ±0.0285	0.4282 ±0.0440	0.3482 ±0.0337	0.4809 ±0.0151
Outcome-PG	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2959 ±0.0217	0.3482 ±0.0337	0.3841 ±0.0452
Outcome-Value	0.2723 ±0.0316	0.2836 ±0.0312	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.4005 ±0.0464
Outcome-RLOO	0.2723 ±0.0316	0.2795 ±0.0352	0.2877 ±0.0285	0.2954 ±0.0209	0.3482 ±0.0337	0.3800 ±0.0517
Info-RL Neighbor	0.2723 ±0.0316	0.2913 ±0.0233	0.2877 ±0.0285	0.2995 ±0.0162	0.3482 ±0.0337	0.4209 ±0.0330
Unstratified	0.2723 ±0.0316	0.2913 ±0.0233	0.2877 ±0.0285	0.3118 ±0.0277	0.3482 ±0.0337	0.4164 ±0.0327
Unweighted	0.2723 ±0.0316	0.3326 ±0.0377	0.2877 ±0.0285	0.3621 ±0.0408	0.3482 ±0.0337	0.4560 ±0.0287
SPAR	0.2723 ±0.0316	0.4000 ±0.0506	0.2877 ±0.0285	0.4200 ±0.0386	0.3482 ±0.0337	0.4805 ±0.0154